Tajveer Singh Dhesi

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EDUCATION

University of Chicago — *Ph.D., Computer Science* September 2025 – Present

- Advised by Prof. Nick Feamster (NOISE Lab). Research areas: web measurement, AI governance, agentic AI privacy, online safety policy.

Colgate University — *B.A., Computer Science and Mathematics (double major)* August 2021 – May 2025

- Graduated *summa cum laude*, with High Honors in Computer Science. Dean's Award with Distinction - 5 consecutive semesters. Laura Sanchis Award for Excellence in Research (2025).

LANGUAGES & TECHNOLOGIES

Programming: Python (Selenium, requests, Playwright), R, Java, JavaScript, C, SQL

ML / AI: OpenAI API, Claude Console, Hugging Face transformers, local LLM inference (Ollama, mlx.lm), privacy-aware prompt engineering, evaluation pipeline design

Systems & tooling: Linux/Unix, Docker, Git, LaTeX, headless Chromium, MCP (Model Context Protocol)

RESEARCH EXPERIENCE

Graduate Researcher — NOISE Lab, University of Chicago *Advisor: Prof. Nick Feamster* September 2025 – Present

Agentic AI Data Leakage Benchmarking March 2026 – Present

- Designing a benchmark suite to measure data leakage in agentic LLM systems on realistic information-relay tasks (email drafting, social posts) where inputs mix relevant and irrelevant sensitive content; the benchmark quantifies how often agents leak the irrelevant portion despite instructions to use only the relevant subset.
- Spanning open-source (7B–32B) and frontier closed-source agents to enable utility-vs-leakage comparisons across enterprise and consumer deployment tiers.
- Evaluating candidate defenses against the benchmark to characterize the utility-to-defense-efficacy trade-off - directly relevant to NIST AI RMF guidance and emerging FTC and state-AG enforcement around AI privacy harms.

Transformer-Based Web Scraping for Age-Gate Detection September 2025 – Present

- Extending prior thesis work to detect age gates at deeper purchase-flow interaction points (cart, checkout, account creation) using a transformer-based scraping agent that navigates without site-specific heuristics — motivated by recent mid-flow verification mandates (UK LCCP 2025, NSW/Queensland delivery rules).

Senior Thesis — Colgate University *"Measuring the Variety of Website Arrival Age Gates"* August 2024 – May 2025

- Built an automated pipeline (Selenium + headless Chromium + GPT-4o-mini classifiers) measuring site-arrival age gates across 4,985 international websites in the e-cigarette, alcohol, and gambling industries; cataloged 595 gates across 8 ccTLDs and found virtually all rely on trivial self-attestation.
- Translated findings into policy framing tied to specific jurisdictional regimes (Canada's Vaping Promotion Regulations, UK Gambling Commission LCCP, NSW alcohol delivery rules).

PUBLICATIONS

- T. S. Dhesi, N. Apthorpe. *"Measuring the Prevalence and Variety of Online Age-Gates."* ConPro 2025.
- T. S. Dhesi, A. W. Bannister, A. Manzourolajdad. *"Deep Learning of Ligand-bound RNA Tertiary Structures Diverges from Learning Unbound Ones."* bioRxiv, 2023. doi:10.1101/2023.09.13.557627.

TEACHING & SELECTED PROJECTS

Teaching Assistant, University of Chicago September 2025 – Present

- *Introduction to Computer Security* (TOCTOU exploits, RSA padding-oracle attacks, web-scraping tasks); *Machine Learning for Computer Systems* (network packet-trace ML); *Creating Interactive Systems with User-Centered Design* (mentored 3 student groups on quarter-long HCI projects).

Teaching Assistant, Colgate University August 2023 – December 2023

- Supported 50+ students across two sections of Discrete Structures (COSC 290) and one section of Data Structures & Algorithms (COSC 202), both taught in Java.